



KNOWLEDGE

INSTITUTE OF TECHNOLOGY, SALEM

AICTE, uGC Approved | Anna university Affiliated

An Autonomous Institution

Project Title :	PASSIVE COOLING FOR HOMES		
Team Details			
Team Name :	ECO-COOL		
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Problem Identified

In tropical and high-temperature regions, residential buildings experience excessive indoor heat gain due to high solar radiation, poor thermal insulation, and inadequate natural ventilation. As a result, occupants rely heavily on energy-intensive mechanical cooling systems such as air conditioners and electric fans, leading to increased electricity consumption, higher operating costs, frequent power demand peaks, and adverse environmental impacts. There is a lack of affordable, energy-efficient, and sustainable cooling solutions that can maintain indoor thermal comfort without continuous electrical energy usage.

Existing Solution/ Survey

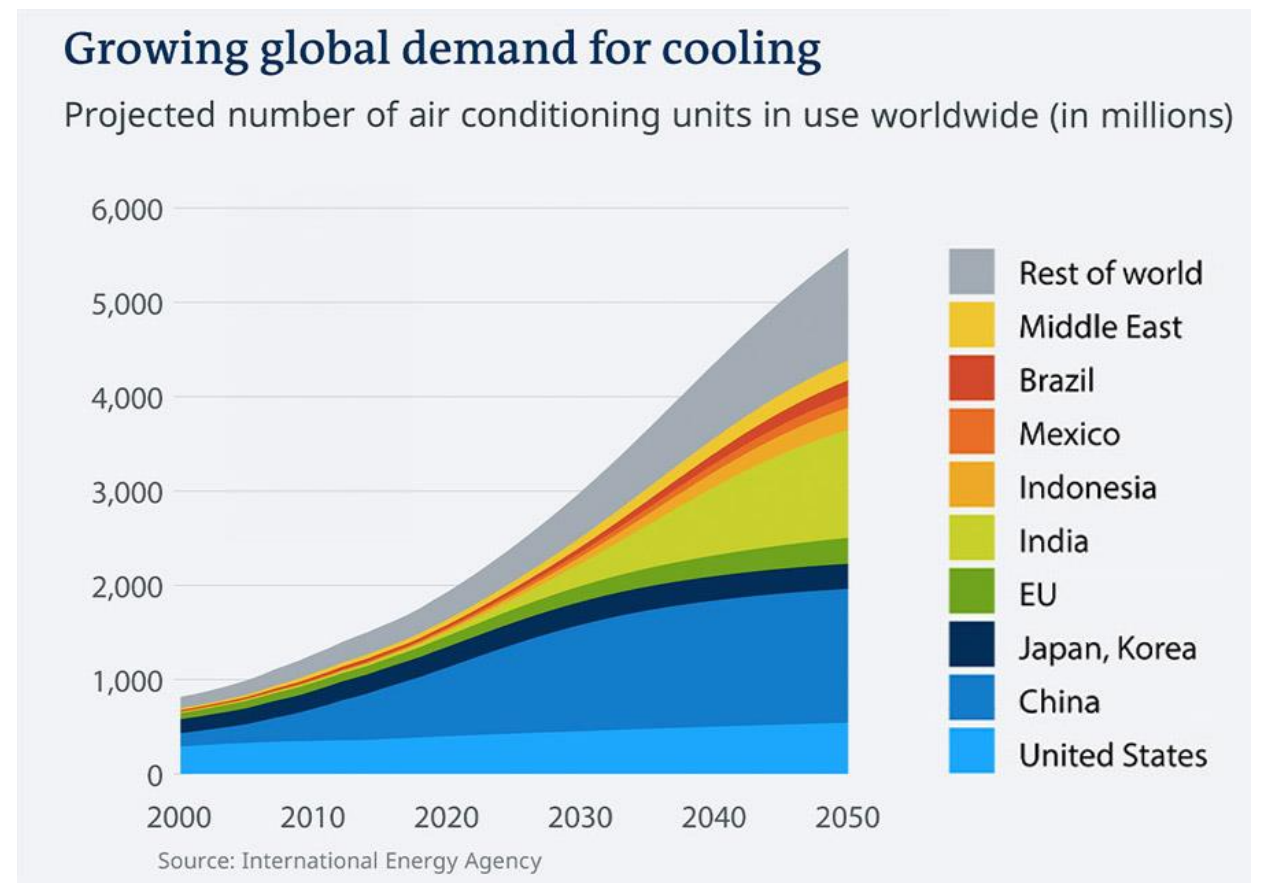
- **Air Conditioners** — High energy consumption, expensive, increases carbon emissions
- **Cool Roof Coatings** — Effectiveness reduces over time, limited impact in humid climates
- **Natural Ventilation** — Depends on wind and weather, inconsistent cooling
- **Evaporative Cooling** — Ineffective in high humidity, requires maintenance
- **Passive Design Features** — Difficult and costly to retrofit in existing homes

Sources:

Google Scholar

Notebook LM

ScienceDirect — Passive Cooling Studies



Proposed Solution

The proposed system integrates low-cost passive cooling techniques such as **reflective roofing**, **thermal insulation**, and enhanced **natural ventilation** to reduce indoor heat gain in residential buildings. An **ESP32-based temperature** monitoring system with indoor and outdoor sensors continuously records temperature data at **one-minute intervals**, enabling real-time performance evaluation of the passive cooling approach.

Novelty of the Solution:

- Combines **passive cooling design** with **real-time monitoring**
- Low-cost, scalable, and suitable for **existing homes**
- Enables **data-driven comparison** of indoor vs outdoor thermal behavior



TESTING DATA

	A	B	C	D	E
1	Date	Time	t1	t2	
2	2026-01-07	14:52:27	29.25	29.12	
3	2026-01-07	14:53:00	29.31	29.06	
4	2026-01-07	14:54:00	29.31	29.75	
5	2026-01-07	14:55:00	30.06	33.31	
6	2026-01-07	14:56:00	31.31	35.75	
7	2026-01-07	14:57:00	32.62	37.5	
8	2026-01-07	14:58:00	33.81	39.12	
9	2026-01-07	14:59:00	34.88	40.19	
10	2026-01-07	15:00:00	35.62	40.88	
11	2026-01-07	15:01:00	36.12	40.88	
12	2026-01-07	15:02:00	36.5	41.25	
13	2026-01-07	15:03:00	36.12	42	
14	2026-01-07	15:04:00	36.5	42.06	
15	2026-01-07	15:05:00	36.81	41.69	
16	2026-01-07	15:06:00	37.06	41.31	
17	2026-01-07	15:07:00	37.38	41.12	
18	2026-01-07	15:08:00	37.81	41.06	
19	2026-01-07	15:09:00	37.94	41.19	
20	2026-01-07	15:10:00	38.19	41.31	
21	2026-01-07	15:11:00	38.5	41.56	
22	2026-01-07	15:12:00	38.56	41.69	
23	2026-01-07	15:13:00	38.44	41.44	
24	2026-01-07	15:14:00	37.88	42.31	
25	2026-01-07	15:15:00	37.06	43	
26	2026-01-07	15:16:00	36.44	43.31	
27	2026-01-07	15:17:00	36.38	42.19	
28	2026-01-07	15:18:00	36.88	41.62	
29	2026-01-07	15:19:00	37.31	41.31	
30	2026-01-07	15:20:00	37.69	41.19	
31	2026-01-07	15:21:00	37.94	41	
32	2026-01-07	15:22:00	37.94	40.62	
33	2026-01-07	15:23:00	38.19	40.75	
34	2026-01-07	15:24:00	38.38	40.75	
35	2026-01-07	15:25:00	38.56	40.94	
36	2026-01-07	15:26:00	38.44	41.06	
37	2026-01-07	15:27:00	38.38	41.75	
38	2026-01-07	15:28:00	38.12	41.75	
39	2026-01-07	15:29:00	38.06	41.44	
40	2026-01-07	15:30:00	38.12	41.19	



BLOCK DIAGRAM

ENVIRONMENTAL HEAT



PASSIVE COOLING TECHNIQUES



REDUCED INDOOR HEAT GAIN



INDOOR AND OUTDOOR TEMPERATURE SENSING



ESP32 MICROCONTROLLER



TIME STAMPING



DATA LOGGING



TEMPERATURE ANALYSIS



PERFORMANCE EVALUATION

Technology Stack / Methodology

COMPONENT	PURPOSE	APPROX. COST (INR)
ESP32 Microcontroller	Data processing & control	400
DS18B20 Temperature Sensors (2)	Indoor & outdoor temperature sensing	800
RTC Module (DS3231)	Time stamping	200
SD Card Module + SD Card	Data storage	1300
Jumper Wires & PCB	Connections	100
Power Supply (Adapter/Battery)	System power	1200
Total Hardware Cost		4000

Software Developed

- ESP32 firmware programmed using Arduino IDE
- Code handles temperature acquisition, time stamping, and data logging
- Data stored in CSV format for easy analysis
- No internet dependency required

Cost of Development & Selling Development Cost

- Hardware: 11,550
- Passive cooling materials (prototype): 11,000
- Total Development Cost: 12,500

Estimated Selling Cost

- Manufacturing + installation + profit margin
- Estimated Market Price: 14,000 ₹5,000 per unit

Impacts

Technical Impact

- Reduces indoor temperature by 5-7 °C
- Lowers energy consumption without active cooling
- Enables real-time performance monitoring

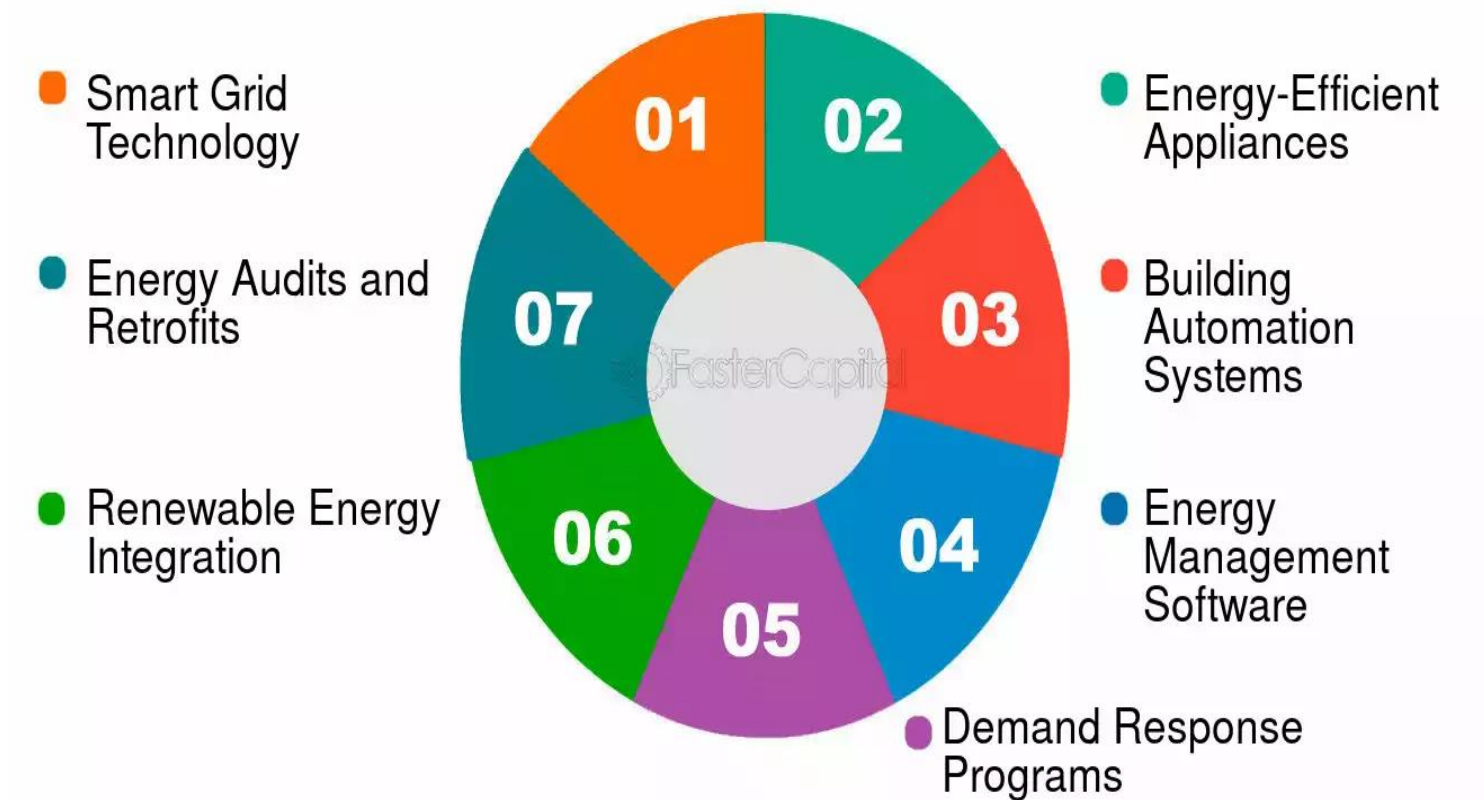
Social & Economic Impact

- Reduces electricity bills
- Affordable for rural and low-income homes
- Environment-friendly solution

Benefits to Users / Industry

- Improved indoor comfort
- Low cost and easy to implement
- Promotes sustainable housing practices

Exploring innovative ways to reduce energy consumption through technology



CONCLUSION

The innovation presents a low-cost passive cooling system for homes that **reduces indoor temperature** by **5-7 °C** without using energy-intensive cooling devices. By combining simple passive cooling techniques with an ESP32-based real-time **temperature monitoring system**, the solution provides measurable, data-driven validation of thermal performance. The system is affordable, easy to retrofit in **existing homes, and suitable for tropical and energy-constrained regions.**

